Shih Hsun Wang

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Summary

Hello, I am Shih Hsun (Michael), an engineer with an interdisciplinary background in **IT (Information Technology)** and **OT (Operational Technology)**. I specialize in practically **applying AI technology and data science** on manufacturing floors. Inspired by my father, who worked in programming, and having been involved with LEGO robotics (EV3) since elementary school through to university, I have a **passion for the integration of mechanics and programming**. This passion not only led me to choose mechanical engineering for a solid hardware foundation but also to pursue a master's in computer science to delve into **AI** and **data analytics**. I firmly believe that the dual advantage of understanding hardware principles and software development is key to advancing modern smart manufacturing and solving complex production line issues.

Skill

Professional Skills Expert

Professional tools and theoretical knowledge

SolidWorks | Creo (ProE) | 8D Report | RCA | 6 sigma | FMEA | SPC | QC 7 Tools | DOE | YOLO | FastAPI | Python | Langchain

Professional Certificates Expert

Professional qualifications and practice certifications

CQT | Six Sigma(GB) | NVIDIA Jetson (Edge AI) | NVIDIA DLI Certs

Work Experience Skills Expert

Concrete achievements and problem-solving experience

Visual Inspection Equipment Implementation | Cross-border Project Management | Equipment Development | Process Optimization | Smart Manufacturing | IoT Promotion | Standardized SOP | Automated Quality Inspection System Development

Job History

2021-10 ~ present

Process and Equipment Engineer TPI Tung Pei Industrial (Taiwan's largest bearing manufacturer)

2024-10 ~ present

- **Plant-wide Visual Inspection Equipment Implementation and Cost Reduction (25%)**: Served as the person in charge of defect inspection (AOI+AI) equipment across the entire plant, including **cross-border project management** and **equipment development**. Led the introduction of visual inspection equipment from **defining technical specifications** and functional requirements to final acceptance. Conducted **three overseas factory audits** in Hangzhou to verify mechanical stability and AI model integration performance, ensuring supplier equipment met the plant's **cycle time** and **inspection accuracy**. Responsible for the entire rollout process during implementation including **hardware optimization**, **model parameter tuning**, **model training (YOLO)**, and **SOP standardization**. Provided **technical transfer training** to the maintenance team to ensure long-term stable operation and successfully reduced manpower needs by **25%** on two pilot production lines (**4 to 3 people**).

2025-06 ~ 2026-02

- **Process Optimization and Statistical Validation (A/B Testing)**: Conducted **A/B testing** comparing multi-machine and single-machine cleaning processes. Verified through **data visualization** and **statistical testing** that single-machine cleaning met production line requirements, successfully removing one cleaning device from all production lines, achieving an annual cost saving of **2.2 million TWD**.
- **Smart Manufacturing and IoT Promotion**: Led the digital transformation of cleaning equipment, implementing **predictive maintenance** logic and **data visualization** using existing sensors. Collaborated with the electrical control team to deploy functions on the HMI (Human-Machine Interface), and planned with the automation department to build dashboards in the **MES system** for advanced process analysis.

2024-10 ~ 2025-12

- **Automated Quality Inspection System Development:** Developed multi-feature fusion **time-series data** AI models combining **machine learning (Random Forest)** and **signal processing (DEA, Wavelet, FFT)** for automated classification of bearing vibration defects, achieving standardized inspection criteria and significantly reducing dependence on highly skilled personnel.
- **Internal Corporate RAG Knowledge Base Assistant (PoC):** Built an **internal AI assistant** from scratch using **Python** and **LangChain**. Addressed local language processing challenges by introducing a Chinese-optimized **embedding model**. Integrated **RAG** with **FastAPI** to establish a modular architecture and presented a proof of concept (PoC) for internal demonstration.

2022-01 ~ 2022-06

- **Standardization and Training (SOP): Pioneered plant-wide** development of **standardized SOP pocket operation manuals**. Leveraged past teaching and training experience to create systematic materials, streamlined and concentrated key SOP operation steps, and converted text content into graphical flows, successfully reducing new staff training time by **67%** (**3 months to 1 month**).

Education

2024-09 ~ 2026-01

Yuan Ze University

Department of Computer Science and Information Engineering Master

Thesis: **Surface Defect Detection of Bearings Based on Photometric Stereo Combined with YOLO**
Optimized existing image recognition algorithms to successfully overcome detection bottlenecks for specific defect categories, improving model coverage.

2017-09 ~ 2021-06

Yuan Ze University

Department of Mechanical Engineering Bachelor

Thesis: IoT Smart Home
Integrated **Raspberry Pi** and **Node-RED** for hardware-software interfacing, implemented servo motor access control, **real-time video streaming**, and automated lighting, establishing a complete remote interactive dashboard.

Competition/Award

2024-01

AI CUP 2024 Yuanta Artificial Intelligence Open Challenge

Application of RAG and LLM in Financial Q&A

2016-01

First Tech Challenge Australia International Competition

Top 18 in the World (Robot Competition) Sydney, Australia

2021-01

Yuan Ze University Bachelor's Thesis

First Place - IoT Smart Home

2020-01

Yuan Ze University 2020 Maker Competition

Outstanding Award - Smart Living Group

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